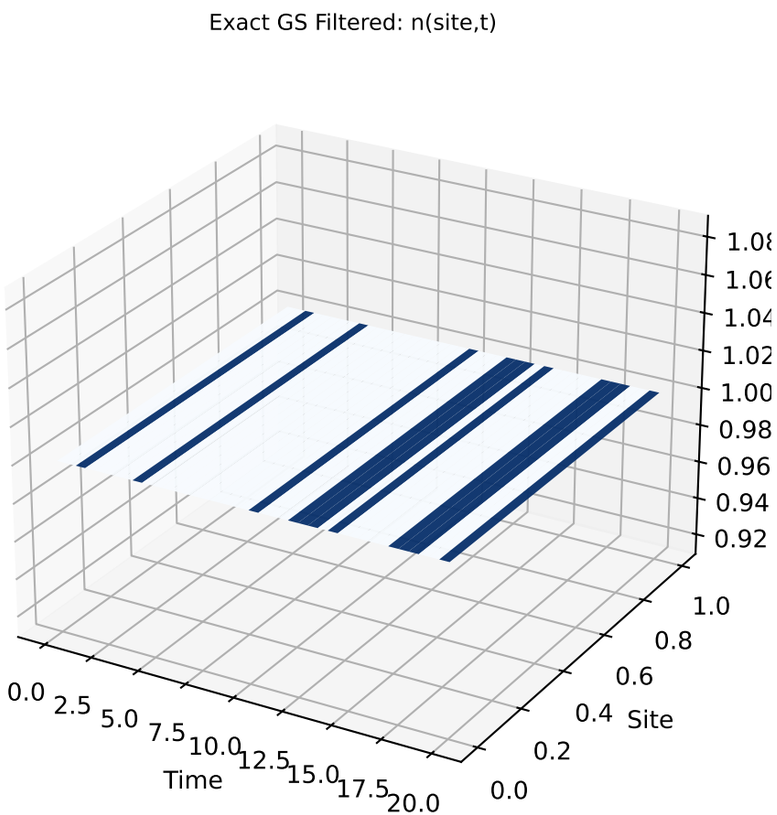
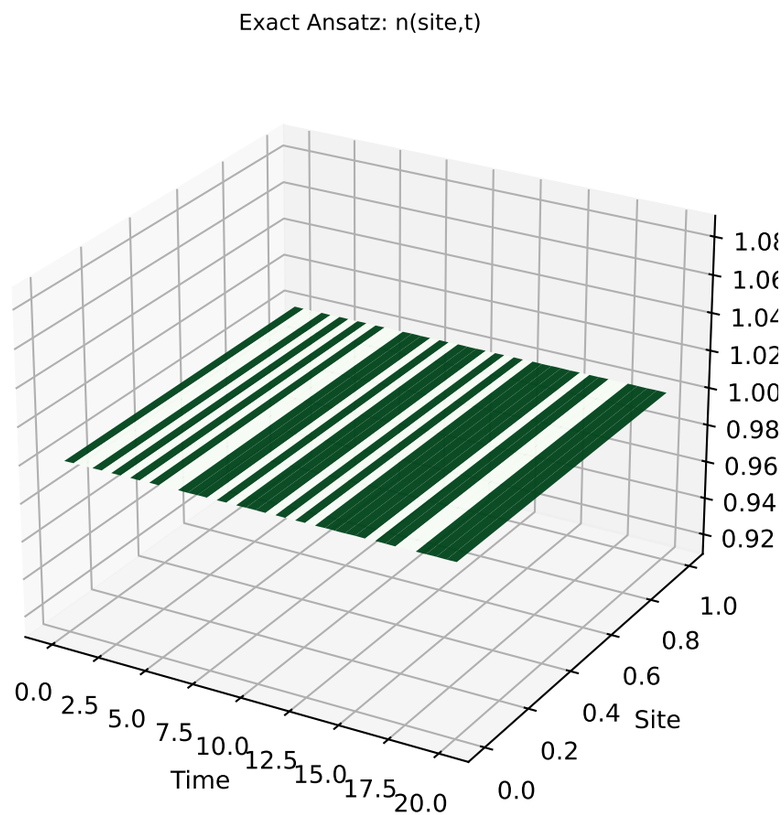
Executed Command

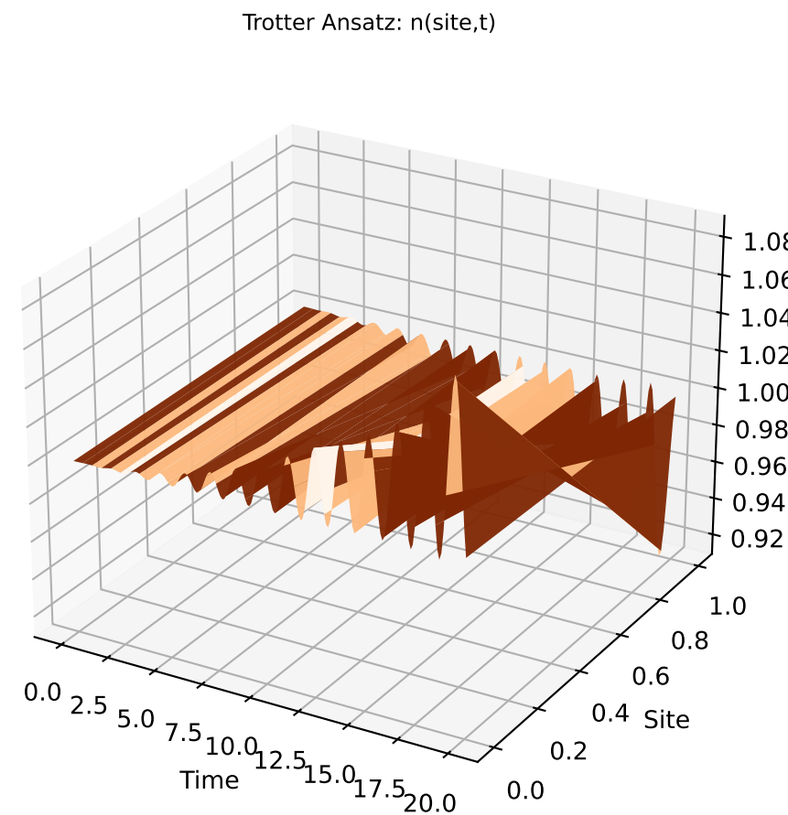
Reference: pipelines/PIPELINE_RUN_GUIDE.md
Script: pipelines/hardcoded_hubbard_pipeline.py

```
/opt/anaconda3/bin/python pipelines/hardcoded_hubbard_pipeline.py --L 2 --t 1.0 --u 4.0 --dv 0.0 --boundary
  periodic --ordering blocked --t-final 20.0 --num-times 201 --suzuki-order 2 --trotter-steps 64 --term-order
  sorted --vqe-reps 2 --vqe-restarts 1 --vqe-maxiter 120 --qpe-eval-qubits 5 --qpe-shots 256 --qpe-seed 11
  --initial-state-source vqe --output-json artifacts/json/H_L2_static_t1.0_U4.0_S64.json --output-pdf
  artifacts/pdf/H_L2_static_t1.0_U4.0_S64.pdf
```

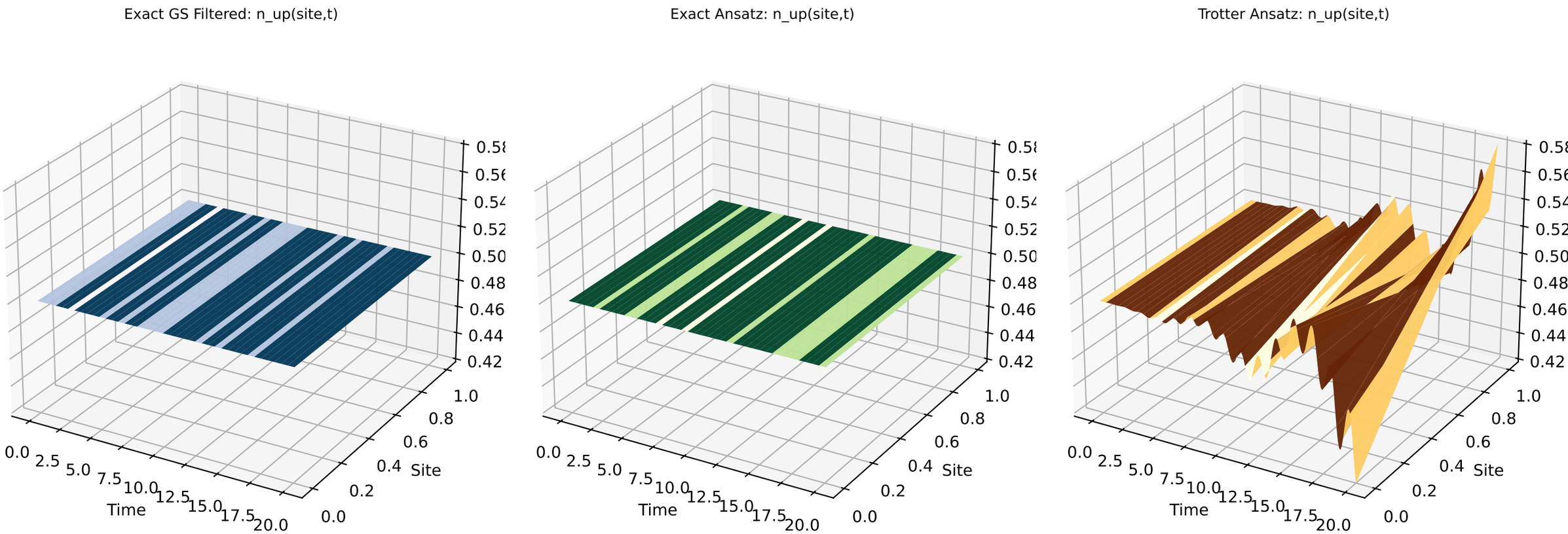
L=2 3D Densities (Total n)



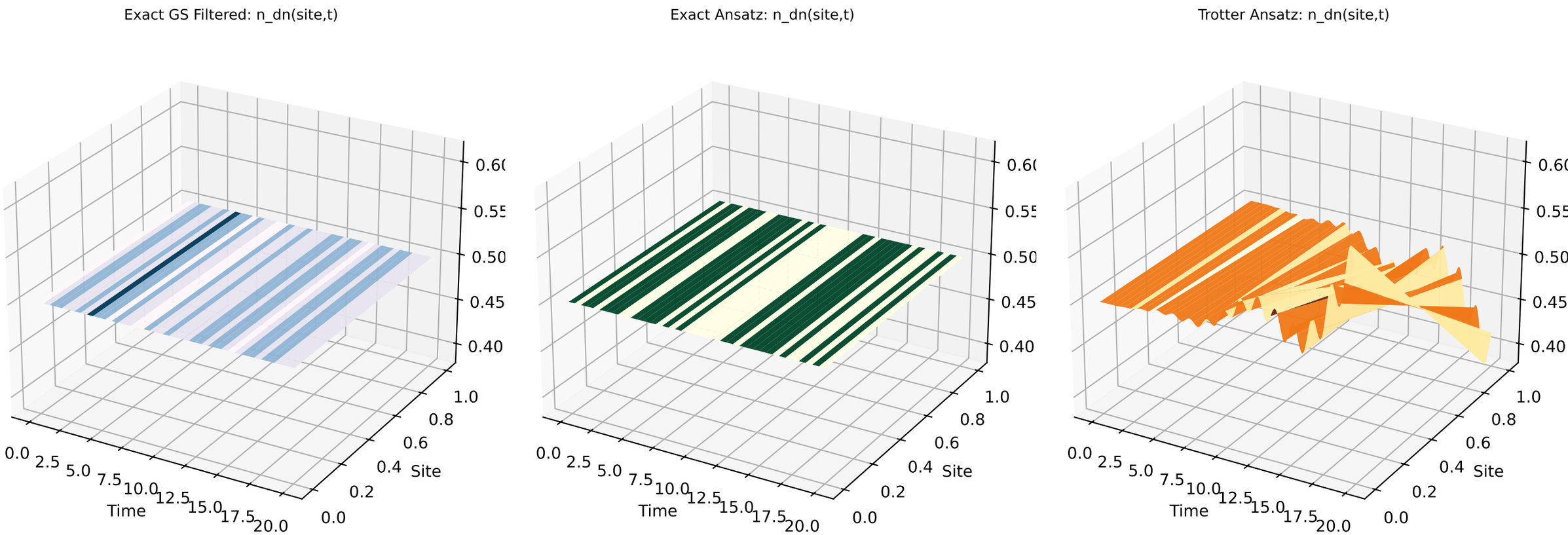




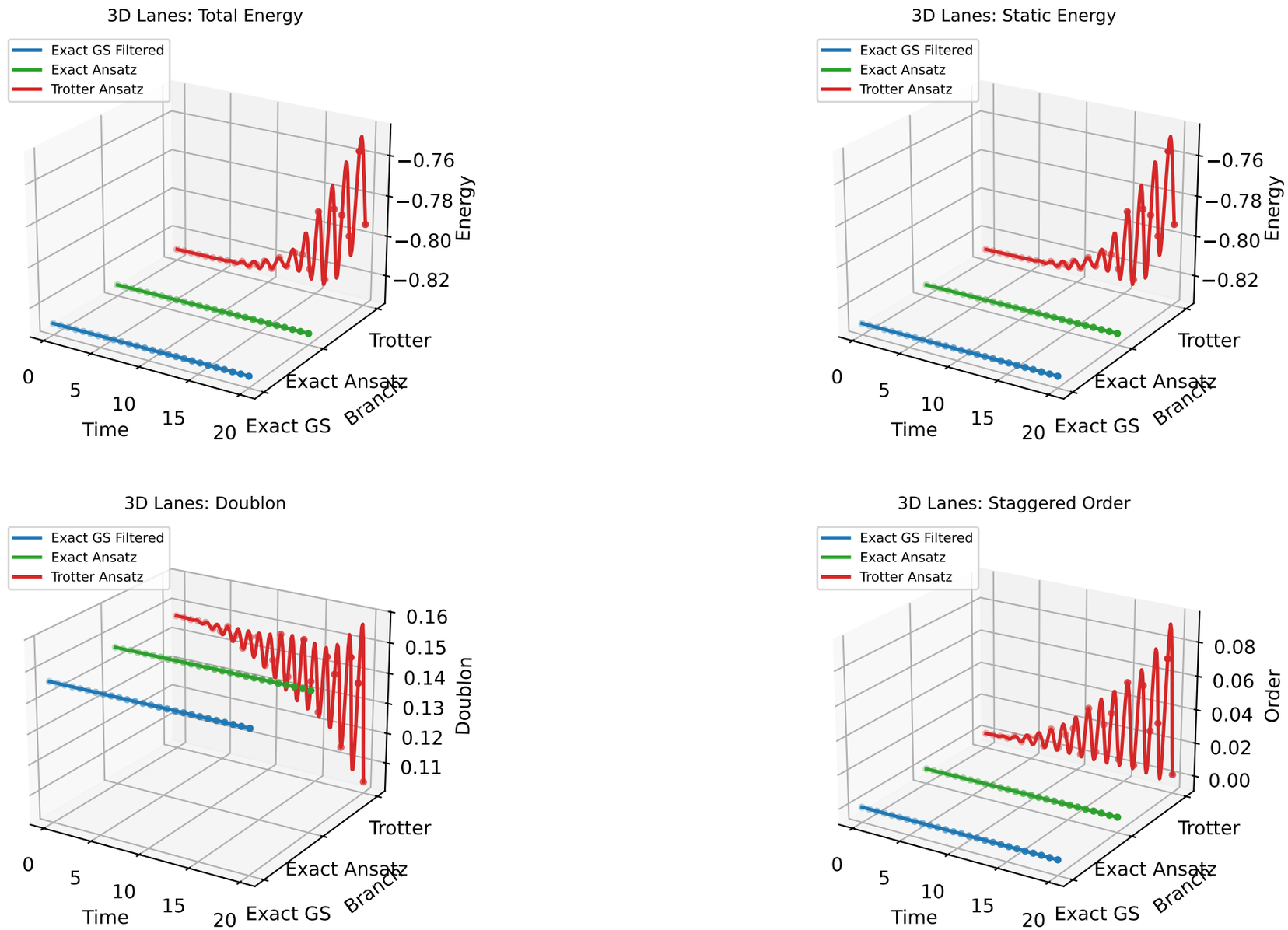
L=2 3D Densities (Spin-Up)

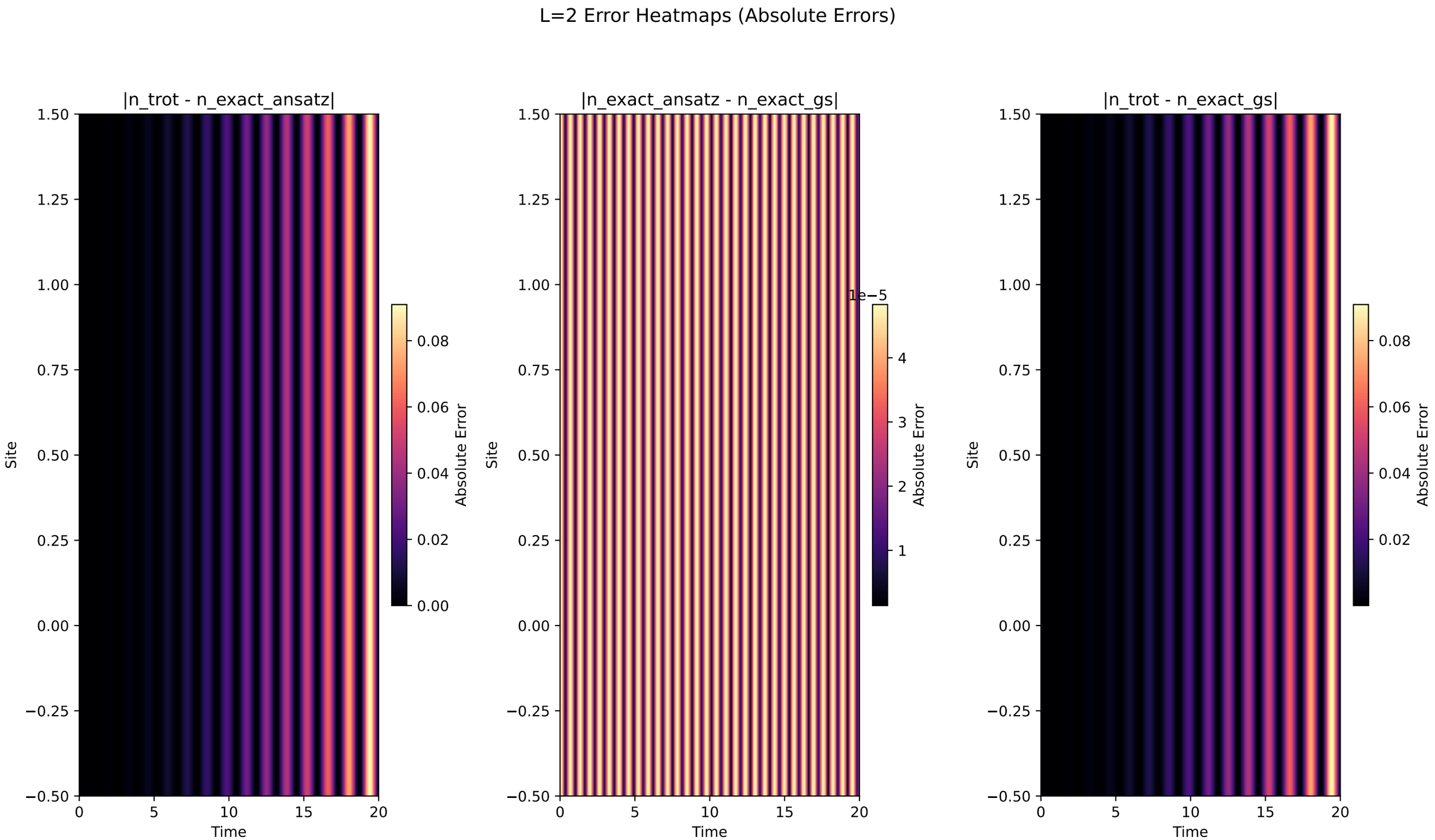


L=2 3D Densities (Spin-Down)

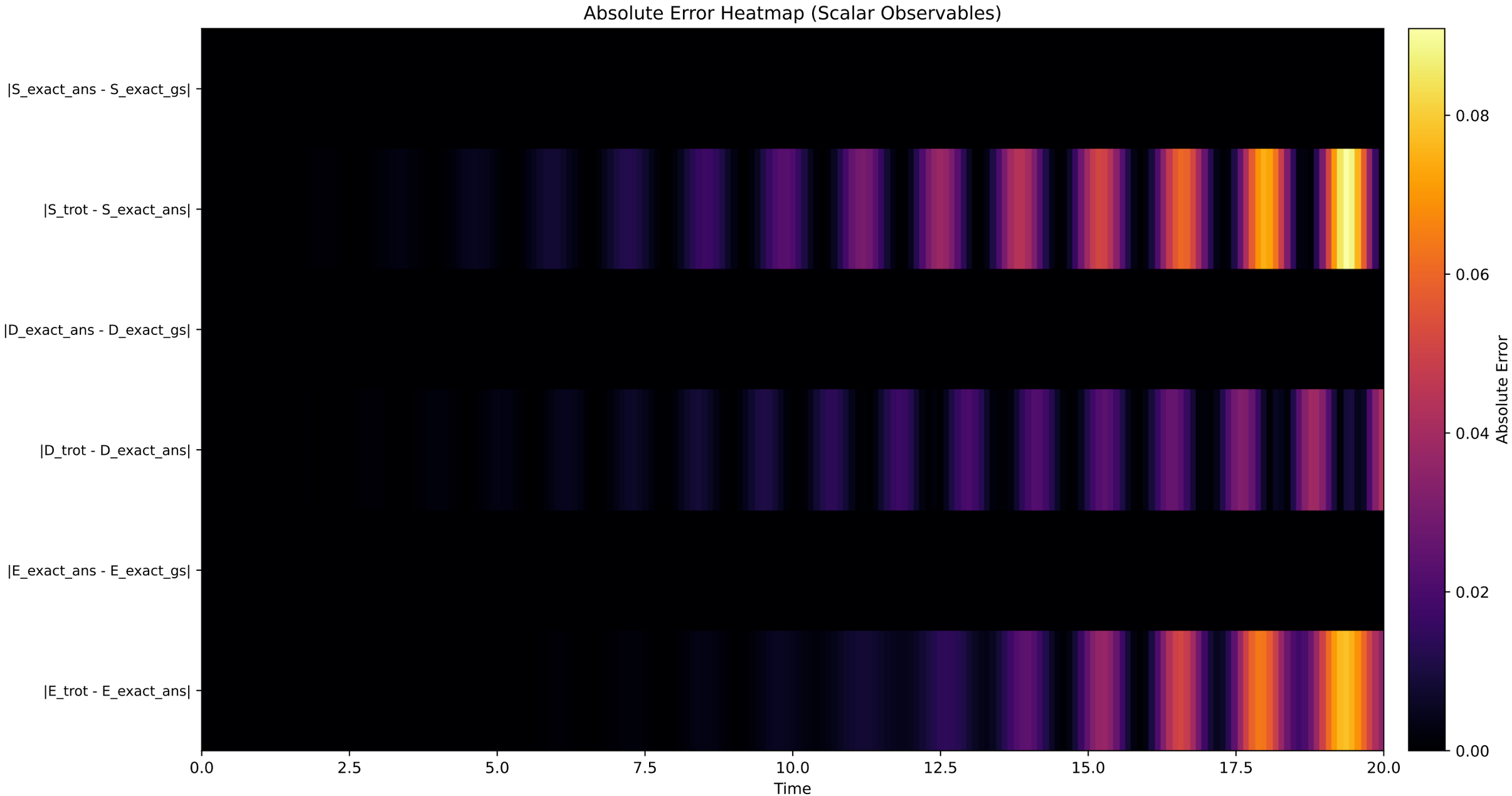


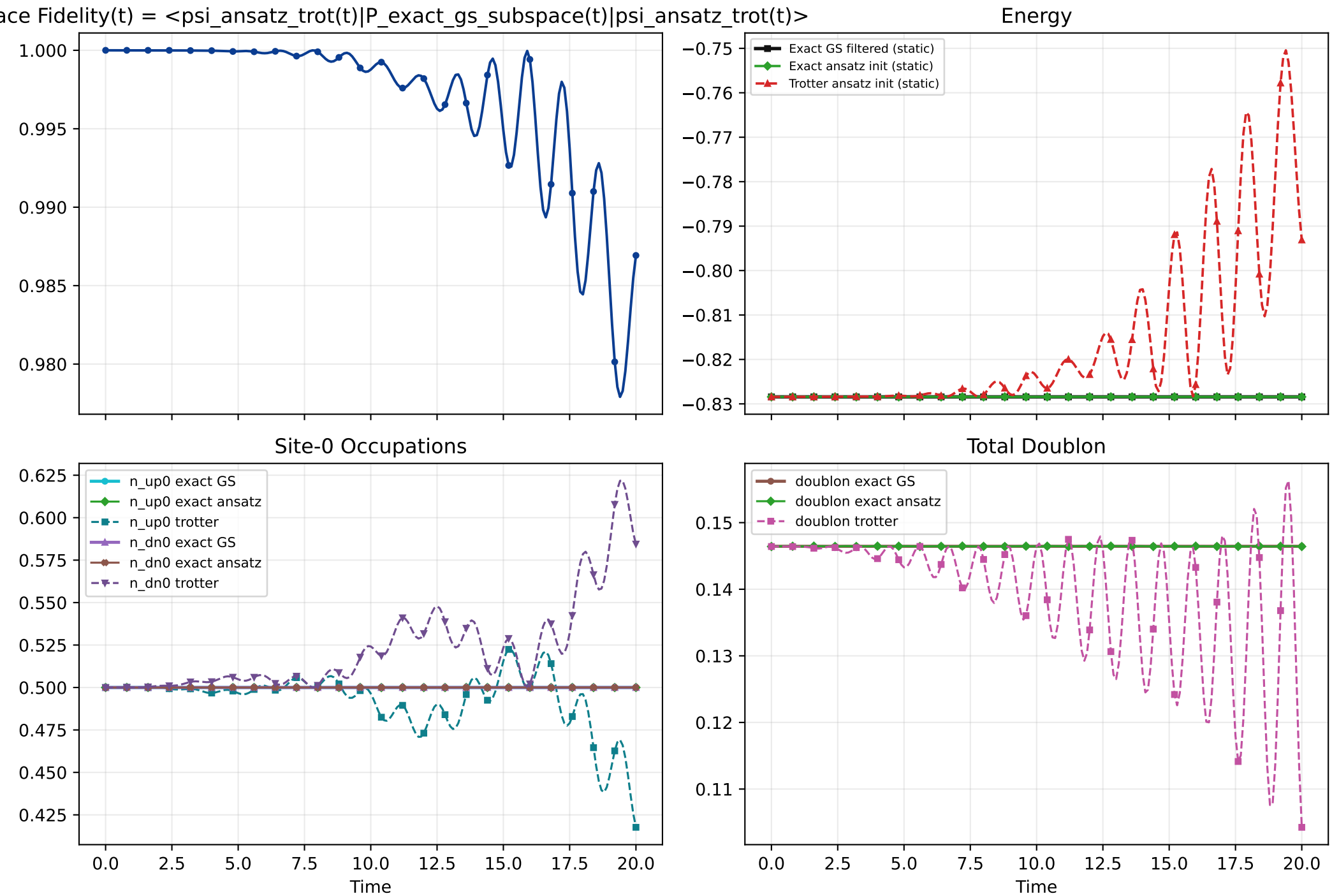
L=2 3D Scalar Observables (Three Evolutions)

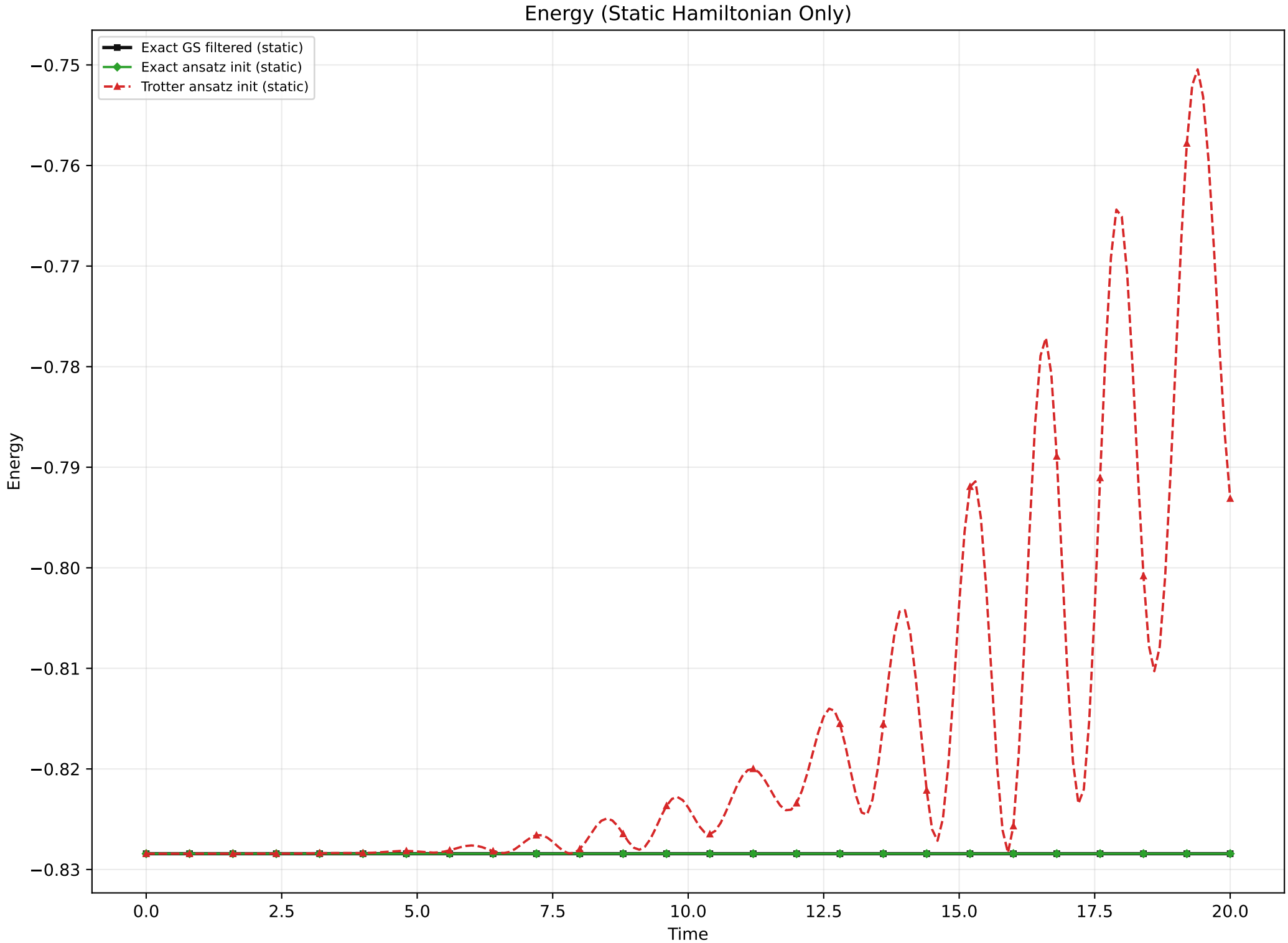


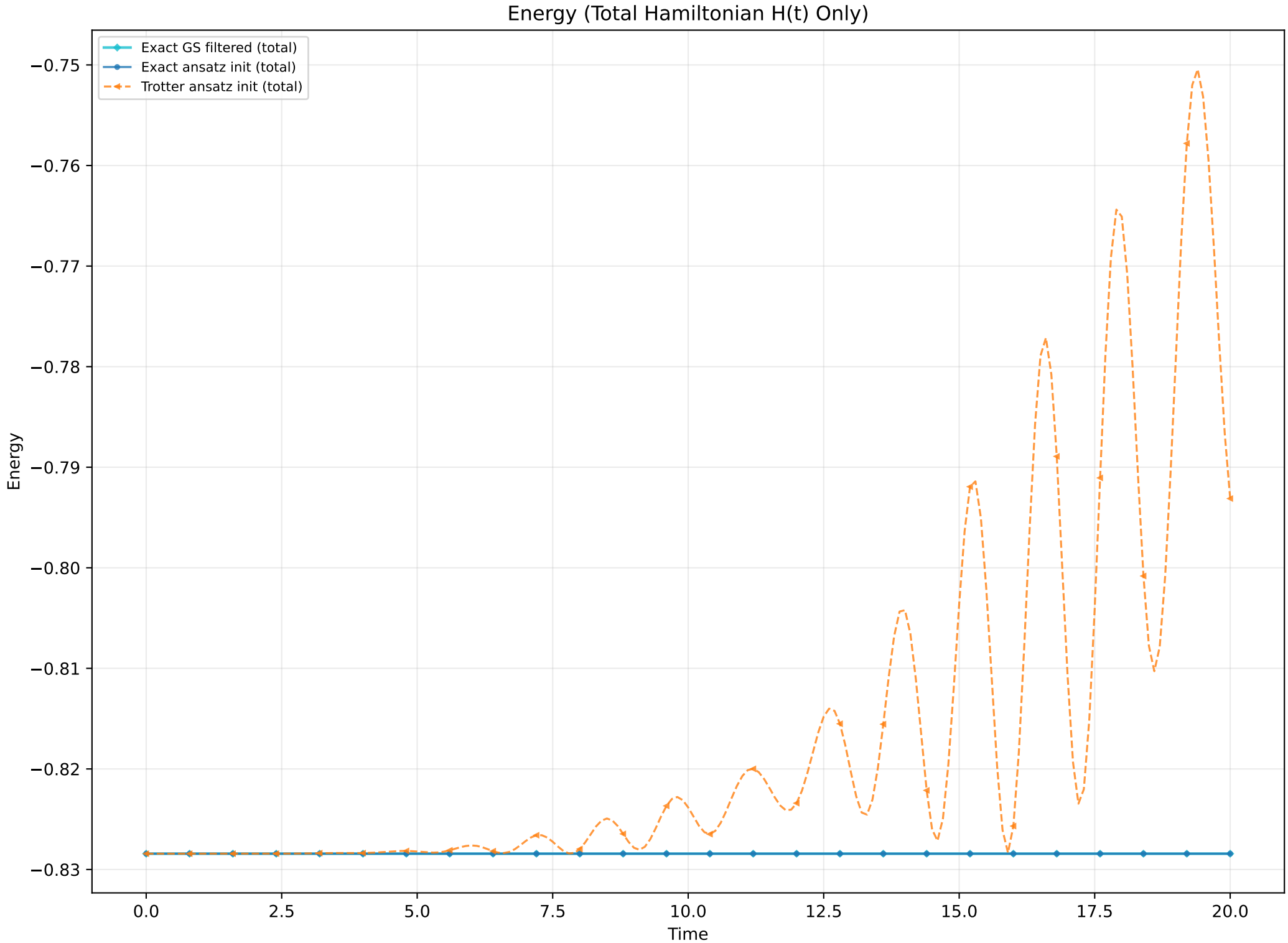

L=2 Error Heatmaps (Absolute Errors)
|n_trot - n_exact_ansatz|
|n_exact_ansatz - n_exact_gs|
|n_trot - n_exact_gs|
Site
Time
Absolute Error
1e−5
1.50
1.25
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
0
5
10
15
20
0.00
0.02
0.04
0.06
0.08
1
2
3
4

L=2 Scalar Error Heatmap




Hardcoded Hubbard Pipeline: L=2
ace Fidelity(t) = <psi_ansatz_trot(t)|P_exact_gs_subspace(t)|psi_ansatz_trot(t)>
Energy
Exact GS filtered (static)
Exact ansatz init (static)
Trotter ansatz init (static)
Site-0 Occupations
n_up0 exact GS
n_up0 exact ansatz
n_up0 trotter
n_dn0 exact GS
n_dn0 exact ansatz
n_dn0 trotter
Total Doublon
doublon exact GS
doublon exact ansatz
doublon trotter
Time
Time

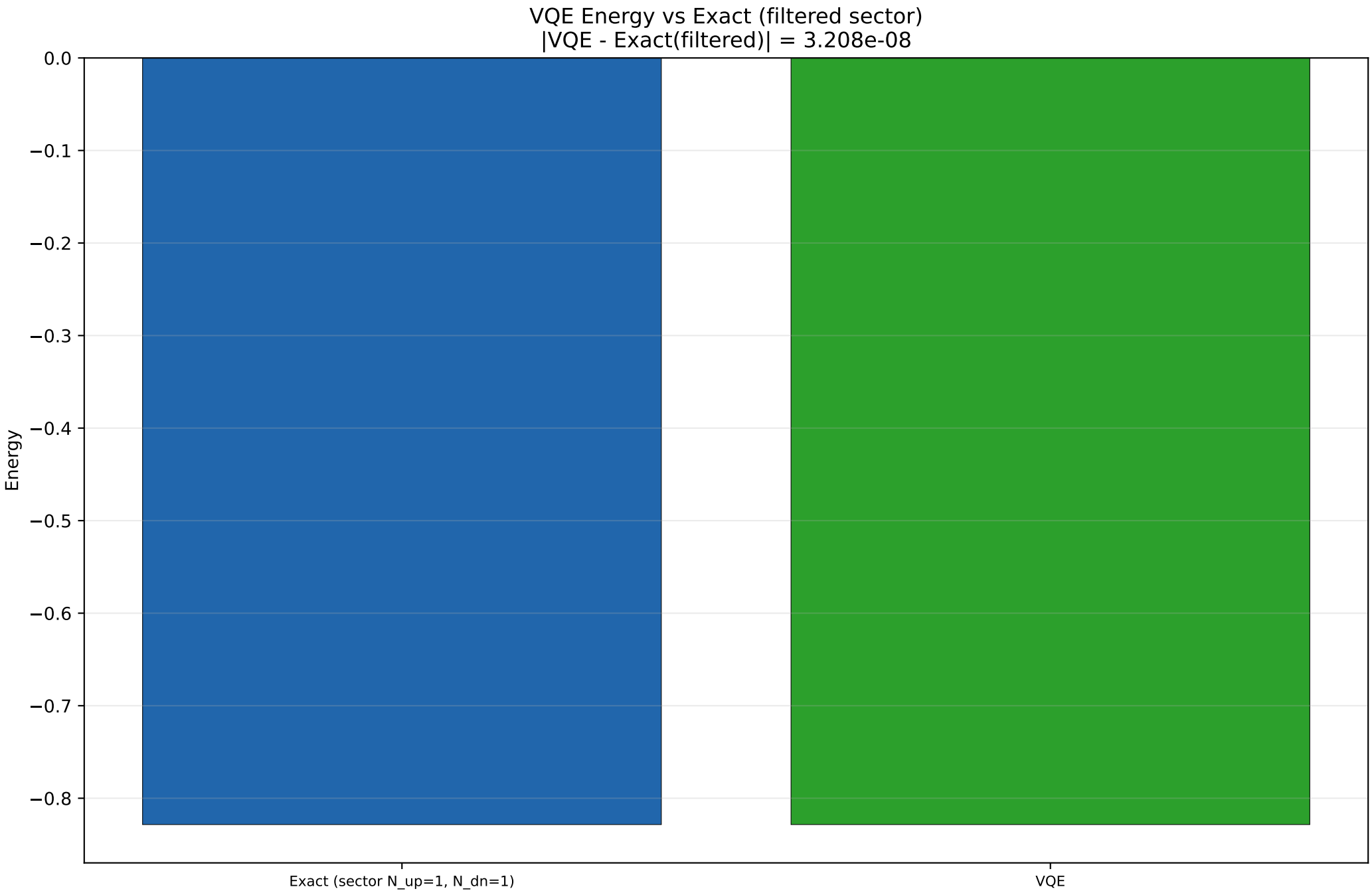

Energy (Static Hamiltonian Only)
Exact GS filtered (static)
Exact ansatz init (static)
Trotter ansatz init (static)
−0.75
−0.76
−0.77
−0.78
−0.79
−0.80
−0.81
−0.82
−0.83
Energy
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time


Energy (Total Hamiltonian H(t) Only)
Exact GS filtered (total)
Exact ansatz init (total)
Trotter ansatz init (total)
Energy
Time
−0.75
−0.76
−0.77
−0.78
−0.79
−0.80
−0.81
−0.82
−0.83
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0

Appendix — L=2

This section contains supporting/redundant views and metadata.
Core non-redundant analysis appears earlier in the PDF.

VQE optimises within the half-filled sector; exact (filtered) is the true sector ground state.
Full-Hilbert exact energy is in the JSON text summary only.



```
Hardcoded Hubbard pipeline summary

settings: {"L": 2, "t": 1.0, "u": 4.0, "dv": 0.0, "boundary": "periodic", "ordering": "blocked", "t_final": 20.0, "num_times":
exact_trajectory_label: Exact_Hardcode
exact_trajectory_method: python_matrix_eigendecomposition
fidelity_definition: fidelity(t) = <psi_ansatz_trot(t)|P_exact_gs_subspace(t)|psi_ansatz_trot(t)>, where P_exact_gs_subspace(t
subspace_fidelity_at_t0: 0.9999999932870716
energy_t0_exact_gs: -0.8284271247461902
energy_t0_exact_ansatz: -0.8284270926619848
energy_t0_trotter: -0.828427092661985
ground_state_exact_energy (full Hilbert): -1.000000000000
ground_state_exact_energy_filtered: -0.8284271247461901
filtered_sector: {'n_up': 1, 'n_dn': 1}
vqe_energy: -0.8284270926619854
qpe_energy_estimate: -0.625
initial_state_source: vqe
hamiltonian_terms: 11
reference_sanity: {'checked': False, 'reason': 'no matching bundled reference for these settings'}
```